

7. SOFTWARE AND HARDWARE REQUIREMENTS

To implement the **BLE-Based Employee Tracking System**, both hardware and software components are required. These components collectively enable real-time data transmission, processing, and visualization.

7.1 SOFTWARE REQUIREMENTS

1. **Arduino IDE**

Used for programming the ESP32 microcontroller with the necessary firmware to detect BLE signals and handle Wi-Fi communication.

2. **Firebase (Google Backend-as-a-Service)**

Cloud database to store employee entry and exit timestamps. Offers real-time data synchronization and scalability.



3. **HTML, CSS, JavaScript (React/Java)**

Used to develop the web dashboard interface to view attendance logs, calculate work hours, and manage employee data.

4. **Firebase Admin SDK / REST APIs**

Enables secure interaction between the web interface and the Firebase backend for reading and writing data.

5. **Google Chrome / Any Modern Web Browser**

Required for accessing the web dashboard.

6. **Operating System**

- For development: Windows 10 or higher / Linux (Ubuntu)
- For deployment: Can be hosted on Firebase Hosting or any compatible cloud server.